10 (a) Table 10.1 shows data for planets A, B, C and D.

Table 10.1

planet	surface temperature / K	diameter / km	gravitational field strength N / kg
A	153	143 000	23.0
B	623	4 800	3.7
C	53	50 000	11.0
D	93	120 000	9.0

(i) State and explain which one of these planets is closest to the Sun.

.....

..... [2]

(ii) Calculate the surface temperature of planet D in °C.

surface temperature = ... °C [1]

(iii) An object falls through a height of 2 m on each of the planets in Table 10.1.

State and explain on which planet the object falls 2 m in the shortest time. Ignore any effect due to the atmosphere.

.....

..... [2]

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(b) Fig. 10.1 shows some of the stages in the life cycle of a massive star.

massive star expands

↓

.....

↓

supernova

↓

nebula which contains .....

↓

leaves behind

..... or .....

Fig. 10.1

(i) Complete Fig. 10.1 by adding the correct terms. [4]

(ii) State the quantity that the brightness of a supernova in a galaxy can be used to determine.

..... [1]

[Total: 10]

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